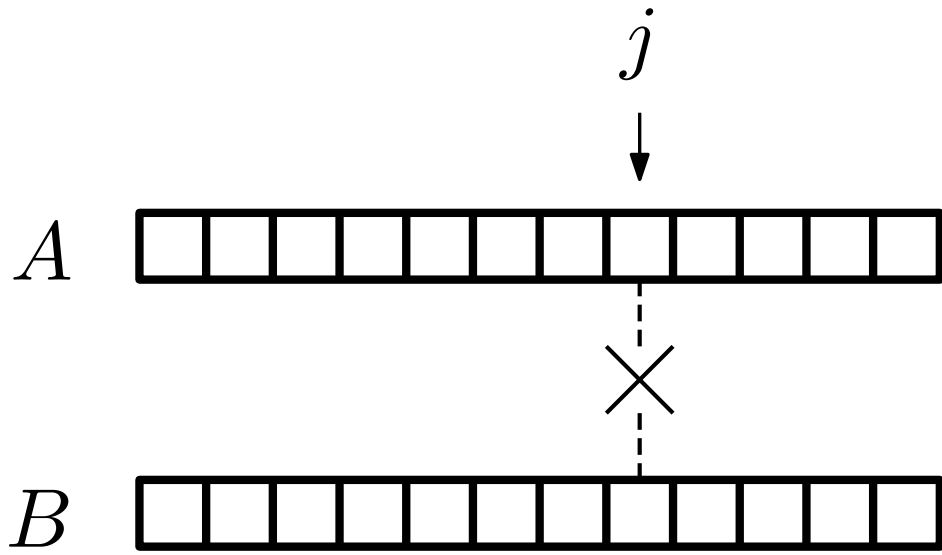$$f_1(A, B) = j$$